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SECTION 1

PHASE 4: FINAL PROJECT REPORT

PROJECT TITLE:

**Business Intelligence Analysis of Brazilian E-Commerce Customer Purchasing Behavior Using
Olist Datasets**

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TABLE OF CONTENTS

1.0 Problem Background	4
1.1 Business Context	4
1.2 Importance of the problem	4
1.3 Expected Impact	5
2.0 Objectives	5
3.0 Dataset Description	6
3.1 Dataset 1 – olist_customers_dataset.csv	6
3.2 Dataset 2 – olist_orders_dataset.csv	6
3.3 Dataset 3 - olist_order_items_dataset.csv	7
3.4 Dataset 4 - olist_products_dataset.csv	7
4.0 Proposed BI Approach	7
4.1 Tools Used	7
4.2 Data Integration, Cleaning, and Transformation Methods	8
4.3 Planned Analysis	9
4.4 Proposed Visualizations	10
5.0 Data Workflow	10
5.1 Data Input	10
5.2 Data Cleaning	11
5.3 Data Transformation	11
5.4 Data Integration and Join Process	12
5.5 Analytical Outputs	13
5.5.1 Product Category Revenue Analysis	13
5.5.2 Regional Customer and Revenue Analysis	13
5.5.3 Monthly Sales Trend Analysis	14
5.5.4 Forecasting Analysis Using ARIMA and ETS	14
5.5.5 Delivery Performance Analysis	16
5.5.6 Customer Spending Segmentation	16
6.0 Challenges Faced	17

7.0 Dashboard Overview	18
7.1 Dashboard 1: Sales & Product Performance Dashboard.....	18
8.0 Key Insight.....	23
8.1 Revenue Performance & Growth Trends	23
8.2 Product Category Performance & Market Concentration	23
8.3 Geographic Revenue Distribution & Regional Market Strength	23
8.4 Delivery Performance & Logistics Efficiency.....	23
8.5 Customer Segmentation & Spending Behavior.....	24
9.0 Business Recommendations	24
9.1 Optimize Delivery Performance & Reduce Late Orders	24
9.2 Expand High-Performing Product Categories	24
9.3 Geographic Market Expansion & Regional Development	24
9.4 Optimize Logistics Costs & Regional Performance	25
9.5 Customer Segmentation Strategy & Retention Focus	25
10.0 Discussions.....	25
10.1 Overall Business Performance	25
10.2 Product and Regional Performance	25
10.3 Logistics and Customer Behavior.....	26
10.4 Limitations and Improvement	26
11.0 Conclusions.....	26
12.0 References	27
13.0 Appendix.....	27

1.0 Problem Background

1.1 Business Context

With the rapid growth of the e-commerce industry, enterprise can now access a large number of transactions and customer data to optimize their operations. Olist, the main department store market for e-commerce in Brazil are playing a crucial role by connecting thousands of small businesses across Brazil with customer. Operating in such large and competitive markets means tracking sales performance, understanding customer behaviour and monitoring logistics efficiency are absolute priorities. In order to convert a large amount of daily raw data in practical business insights, modern companies rely on business intelligence tools. Using Alteryx Designer for data preparation and Microsoft Power BI for analysis, organizations can make strategic, data-driven decision instead of relying on speculation.

1.2 Importance of the problem

Although e-commerce platforms generate a large amount of data every day, this information is often trapped in a separate silo such as isolated customer database, order logs, product details and shipping details. If there is no appropriate BI solution to connect these components, there will be several major operational problems.

First, Brazil's vast geographical environment makes transportation a headache, which are the difference between the expected delivery date and the actual delivery date often occurs and it will directly damage customer satisfaction. Secondly, if product details are not integrated with actual sales data, it is difficult to see which product categories really drive revenue and which product categories cause delivery delays or regional fulfilment problems. Finally, lack of customer segmentation means that the marketing team cannot identify high-spending groups, which lead to general activities that waste the budget. In order to prevent poor business forecasts and missed regional opportunities, it is obviously necessary to use Alteryx to clean up and model these data, laying a solid foundation for Power BI to track key performance indicators.

1.3 Expected Impact

Building this BI solution will immediately improve market efficiency and revenue growth. By clearly planning delivery delays and shipping schedules in different states, management can reevaluate logistics partnerships and provide customers with more accurate delivery estimates. In terms of products, determining the best performing category and regional preferences will directly help manage inventory and attract suitable merchants to join the platform. In addition, dividing the customer group into clear consumption layers like low, medium, high and tracking their location will allow the marketing team to design highly targeted promotional activities to ensure that the budget is used for most profitable regions and customer segments.

2.0 Objectives

The main goal of this project is to build an end-to-end business intelligence solution that solve the data fragmentation issues within the Olist e-commerce platform. To achieve this, the project will focus on these objectives:

1. To integrate and model four core e-commerce datasets (Customer, Order, Order Items and Products) from the Olist platform into a unified relational data structure using Alteryx.
2. To perform data cleaning and transformation processes, including handling missing values, removing duplicate records, standardizing data-time formats and creating custom analytical fields like `spending_level`.
3. To analyse retail sales trends, shipping fulfilment performance and product category revenue to identify meaningful business insights.
4. To segment the consumer base by geographic location and total purchasing value to provide a clear, structured framework for targeted marketing strategies.
5. To develop interactive dashboards using Microsoft Power BI for data visualization and business decision support.

3.0 Dataset Description

This project uses four datasets from the [Brazilian E-Commerce Public Dataset by Olist](#) available on Kaggle.

3.1 Dataset 1 – olist_customers_dataset.csv

This dataset contains customer information and their geographic location data. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse customer distribution by city and state and understand regional purchasing patterns.

The key variables include:

Variable	Description	Variable	Description
customer_id	Unique customer identifier for each order	customer_city	Customer city name
customer_unique_id	Unique customer identifier	customer_state	Customer state name

3.2 Dataset 2 – olist_orders_dataset.csv

This dataset contains order-level information and order processing timestamps. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse order status, delivery performance, and transaction timelines.

The key variables include:

Variable	Description	Variable	Description
order_id	Unique order identifier	order_purchase_timestamp	Customer purchase timestamp
customer_id	Unique customer identifier for each order	order_delivered_customer_date	Actual order delivery date to customer
order_status	Order status such as delivered, shipped, and cancelled	order_estimated_delivery_date	Estimated order delivery date to customer

3.3 Dataset 3 - olist_order_items_dataset.csv

This dataset contains detailed information about products purchased within each order. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse sales performance, pricing, and shipping costs.

The key variables include:

Variable	Description	Variable	Description
order_id	Unique order identifier	price	Product price
product_id	Unique product identifier	freight_value	Product shipping cost
seller_id	Unique seller identifier		

3.4 Dataset 4 - olist_products_dataset.csv

This dataset contains product-related information and product category details. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse how product categories affect sales performance.

The key variables include:

Variable	Description	Variable	Description
product_id	Unique product identifier	product_category_name	Product category

4.0 Proposed BI Approach

4.1 Tools Used

The main tools used in this project are **Alteryx Designer** and **Microsoft Power BI**.

Alteryx Designer will be used for data preparation and workflow development. It used to import multiple Olist CSV datasets and do multiple functions to produce a cleaned dataset. The data preprocessing flows such as cleaning missing values, removing duplicates, transforming data types, joining related tables, creating calculated fields, and lastly exporting the cleaned dataset.

Microsoft Power BI will be used for data visualization, summarized report and dashboard development. It allows project team to create interactive dashboards, KPI cards, charts, slicers, filters, business reports

and so on. Power BI is mostly suitable for this project because it can present business insights in a visual format that is easy for project team to understand and act as presentation use.

The combination of Alteryx and Power BI supports the full BI process from data preparation to insight generation. For the examples, Alteryx will handle the backend data processing while Power BI will focus on dashboard design and business storytelling such a win-win situation.

4.2 Data Integration, Cleaning, and Transformation Methods

The data preparation process will initiate by importing the selected datasets into **Alteryx Designer**. It will begin by importing four related Olist datasets such as orders, customers, order items, and products. For the example, the Olist dataset contains multiple related CSV datasets that can be joined using keys such as `order_id`, `customer_id`, and `product_id`.

Besides, data integration will be performed at an analytical level instead of treating them as one single database for the datasets from different source. The data analysis will compare common business dimensions such as sales, profit, customer segment, product category, region, purchase behaviour, and marketing response. This approach allows the project to build a meaningful retail BI solution without if all datasets belong to the same organization.

After that, the data cleaning process will also include removing duplicate records, handling missing values, correcting inconsistent data formats, standardizing column names, and removing unnecessary fields. For example, missing product categories will be replaced with "Unknown", and date fields will be converted into proper DateTime format. For instances, the missing product category names and incomplete delivery dates must be handled during data cleaning to ensure accuracy in product performance and customer spending segmentation analysis.

After doing the cleaning process, the data transformation process will be proceeded including forming new fields in datasets such as `revenue`, `delivery_time_days`, `delivery_status`, `order_purchase_year_month`, and `spending_level`. The cleaned and transformed data can be fully used for data analysis and dashboard visualization in Power BI.

After all data preparation process is completed successfully, the cleaned datasets will be exported from Alteryx and imported into **Power BI** for visualization. Now, **Power BI** will then be used to create dashboards that show business performance, customer insights, and marketing effectiveness.

4.3 Planned Analysis

No.	Planned Analysis	Description	Expected Business Insight
1	Order Status Analysis	It is used to the number of orders based on order status.	Understand different order patterns.
2	Sales Trend Analysis	It is used to analyze monthly and yearly revenue using order purchase date.	Identify revenue growth patterns and sales trends over time.
3	Product Category Performance Analysis	It is used to analyze total revenue by product category.	Identify top product categories.
4	Regional Customer and Revenue Analysis	It is used to analyze customers and revenue by customer state.	Identify most customer regions and highest revenue states.
5	Delivery Performance Analysis	It is used to analyze delivery time and delivery status.	Evaluate performance and late deliveries.
6	Customer Spending Segmentation	Group customers into Low, Medium, and High spending levels.	Identify customer groups based on total spending behaviors.

4.4 Proposed Visualizations

The project will use several visual forms in Power BI to present the analysis clearly and effectively.

Visual Form	Purpose
Line chart	It shows monthly or yearly sales trends.
Bar chart	It compares revenue by product category and customer state.
Map visualization	It displays customer distribution or revenue by state.
Donut chart	It shows order status distribution or customer spending segment distribution.
KPI Cards	It shows total revenue, total orders, total customers, and average delivery time.
Tree map	It shows product category contribution to total revenue.
Stacked column chart	It compares delivery status such as On Time and Late.

5.0 Data Workflow

This section explains the complete workflow using Alteryx Designer including data input, cleaning, transformation, integration, and analysis for the 4 datasets.

5.1 Data Input

Figure 5.1.1 shows the olist_orders_dataset.csv, olist_customers_dataset.csv, olist_order_items_dataset.csv, and olist_products_dataset.csv were imported into Alteryx Designer using the Input Data tool.

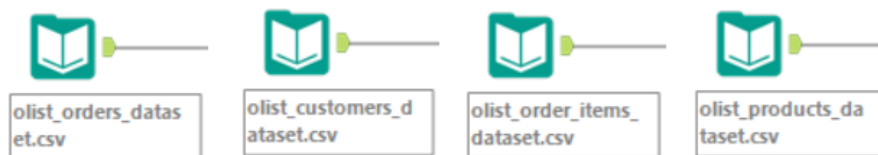


Figure 5.1.1 Data Input Workflow

5.2 Data Cleaning

Figure 5.2.1 shows the data cleaning was performed using Data Cleansing and Unique tools by removing null records, replacing missing numeric values with 0, replacing missing string values with blanks, and trimming whitespace.

For olist_orders_dataset.csv, duplicate records were removed based on the order_id field. For olist_customers_dataset.csv, duplicate records were removed based on the customer_id field. For olist_order_items_dataset.csv, duplicate records were removed based on the order_id and order_item_id field. For olist_products_dataset.csv, duplicate records were removed based on the product_id field.



Figure 5.1.1.1 Data Cleaning Workflow

5.3 Data Transformation

Figure 5.3.1 shows data transformation for olist_orders_dataset.csv was carried out using the Formula and Select tools. The DateTimeParse() function converted timestamp fields into DateTime format, while the Select tool standardized the datatypes of the transformed fields.

Figure 5.3.2 shows data transformation for olist_customers_dataset.csv was carried out using the Select tools. The customer_zip_code_prefix field was deselected as it was not required for the analysis.

Figure 5.3.3 shows data transformation for olist_order_items_dataset.csv was performed using the Select and Formula tools. The price and freight_value fields were converted into FixedDecimal data type, while the shipping_limit_date field was converted into DateTime data type. A new field named revenue was then created by summing the price and freight_value, and converted into FixedDecimal data type.

Figure 5.3.4 shows data transformation for olist_products_dataset.csv was performed using the Select and Formula tools. All fields except product_id and product_category_name were removed as they were not required for the analysis. Lastly, null or empty values in the product_category_name field were replaced with “Unknown” using the Formula tool to improve data completeness and consistency.

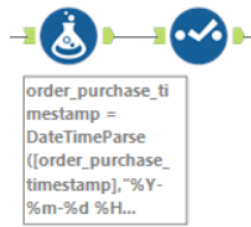


Figure 5.3.1 Data Transformation Workflow for olist_orders_dataset.csv



Figure 5.3.2 Data Transformation Workflow for olist_customers_dataset.csv

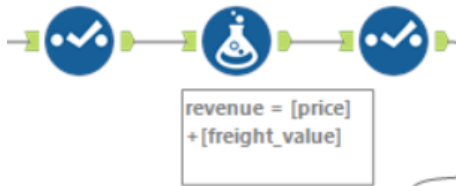


Figure 5.3.3 Data Transformation Workflow for olist_order_items_dataset.csv



Figure 5.3.4 Data Transformation Workflow for olist_products_dataset.csv

5.4 Data Integration and Join Process

Figure 5.4.1 was performed between olist_orders_dataset.csv and olist_customers_dataset.csv using the customer_id. This join allowed each order record to relate to customer location details such as customer city and customer state.

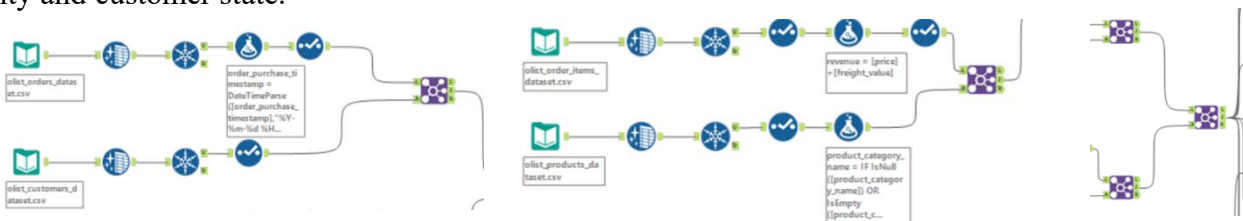


Figure 5.4.1 Data Join 1 & Figure 5.4.2 Data Join 2 & Figure 5.4.3 Final Join

Figure 5.4.2 was joined between olist_order_items_dataset.csv and olist_products_dataset.csv using the product_id. Lastly, figure 5.4.3 was joined using the order_id to combine the order and customer information with the order item and product information. The final joined output provides the main data foundation for product category revenue analysis, regional revenue analysis, monthly sales trend analysis, and forecasting analysis.

5.5 Analytical Outputs

5.5.1 Product Category Revenue Analysis

Product Category Revenue Analysis was used to identify which product categories generated the highest total revenue in the Olist e-commerce dataset. This analysis was performed after the data integration process because some required fields came from different datasets.



Figure 5.5.1 Product Category Revenue Analysis Workflow & Figure 5.5.2 Top 10 Product Categories by Total Revenue Output

In the Alteryx workflow, the dataset was first connected to a Summarize tool. The records were grouped and then a Sort tool was used to arrange the product categories in descending order based on total revenue. A Sample tool was then used to display the Top 10 product categories by total revenue. The output of this analysis helps identify the strongest product categories in terms of sales contribution.

5.5.2 Regional Customer and Revenue Analysis

For revenue analysis, the Sample tool was used to display the Top 10 states with the highest total revenue. For customer analysis, another Summarize tool was used for group records and then the output was then sorted in descending order to identify the Top 10 states with the highest number of customers.



Figure 5.5.3 Regional Customer and Revenue Analysis Workflow

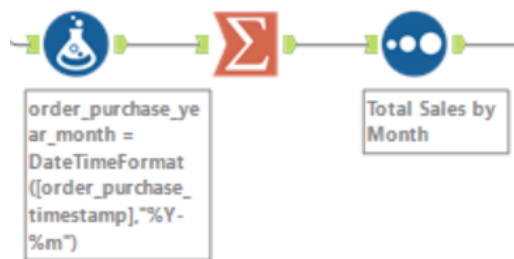
The results from this analysis help the business understand which regions contribute the most revenue and which states have the strongest customer base.

Record	customer_state	Sum_revenue
1	SP	5921678.12
2	RJ	2129681.98
3	MG	1856161.49
4	RS	885826.76
5	PR	800935.44
6	BA	611506.67
7	SC	610213.60
8	DF	353229.44
9	GO	347706.93
10	ES	324801.91

Record	customer_state	CountDistinct_customer_id
1	SP	39981
2	RJ	12303
3	MG	11178
4	RS	5249
5	PR	4840
6	SC	3513
7	BA	3257
8	DF	2062
9	ES	1956
10	GO	1942

Figure 5.5.4 Top 10 States with Highest Total Revenue Output & Figure 5.5.5 Top 10 States with Most Customers Output

5.5.3 Monthly Sales Trend Analysis



Record	order_purchase_year_month	Sum_revenue
1	2016-09	354.75
2	2016-10	56808.84
3	2016-12	19.62
4	2017-01	137188.49
5	2017-02	286280.62
6	2017-03	432048.59
7	2017-04	412422.24
8	2017-05	586190.95
9	2017-06	502963.04
10	2017-07	584971.62
11	2017-08	668204.60
12	2017-09	720398.91

Figure 5.5.6 Monthly Sales Trend Workflow & Figure 5.5.7 Total Sales by Month Output

The monthly revenue output was then sorted in ascending order based on the month field. The final output showed total revenue for each month and became the main input for both sales trend visualization and forecasting analysis. This analysis provides a clear view of revenue changes across months. It can help the business identify periods with stronger sales performance and periods with lower sales activity.

5.5.4 Forecasting Analysis Using ARIMA and ETS

Forecasting Analysis was conducted as an additional analytical technique to estimate future monthly revenue trends. First, a Formula tool was used to create a RevenueLog field using the LOG10 function. Next, another Formula tool was used to create a new field named month_date. After the date conversion, a Sort tool was used to arrange the records in ascending order by month_date. A Filter tool was then applied to keep only complete monthly records from January 2017 to August 2018.

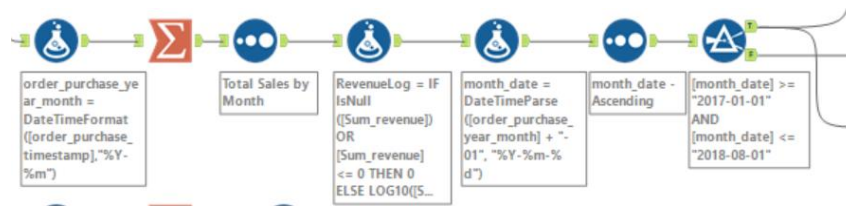
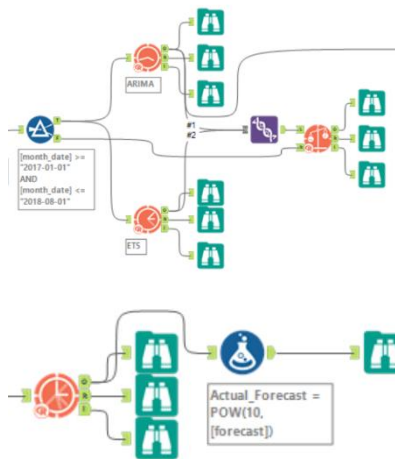


Figure 5.5.8 Forecasting Data Preparation Workflow

After the data preparation steps, both ARIMA and ETS models were applied and compared using the TS Compare tool. Based on the overall accuracy comparison, ARIMA was selected as the final forecasting model.



Record	Model	ME	RMSE	MAE	MPE	MAPE	MASE
1	ARIMA	-3.2837	3.5197	3.2837	-173.5687	173.5687	40.5128
2	ETS	-3.3487	3.5814	3.3487	-176.6078	176.6078	41.3148

Record	Period	Sub_Period	forecast	forecast_high_95	forecast_high_80	forecast_low_80	forecast_low_95	Actual_Forecast
1	2	9	6.000832	6.205895	6.134915	5.86675	5.79577	1001918.74
2	2	10	5.987199	6.324434	6.207705	5.766692	5.649963	970953.80
3	2	11	5.98361	6.529507	6.340553	5.626668	5.437714	962964.69
4	2	12	5.973046	6.728459	6.466984	5.479109	5.217634	939823.36
5	3	1	5.967326	6.972236	6.624401	5.310251	4.962417	927526.46
6	3	2	5.958242	7.224341	6.7861	5.130385	4.692144	908327.55

Figure 5.5.9 ARIMA and ETS Model Comparison Workflow & Result & Figure 5.5.10 Actual Forecast Workflow & Result

The TS Compare output showed that ARIMA achieved a MAPE of approximately 15.98%, while ETS achieved a MAPE of approximately 16.86%. Since ARIMA had the lower MAPE and better overall error performance, it was considered more suitable for the final forecast. The final Actual Forecast output showed a slightly decreasing future monthly revenue trend, with forecast revenue values ranging from approximately 1.02 million to 0.91 million. This indicates that the model predicted a moderate downward trend in future monthly revenue. However, since the historical monthly data range was limited, the forecast result was treated as a supporting analytical output rather than a fully conclusive prediction.

5.5.5 Delivery Performance Analysis

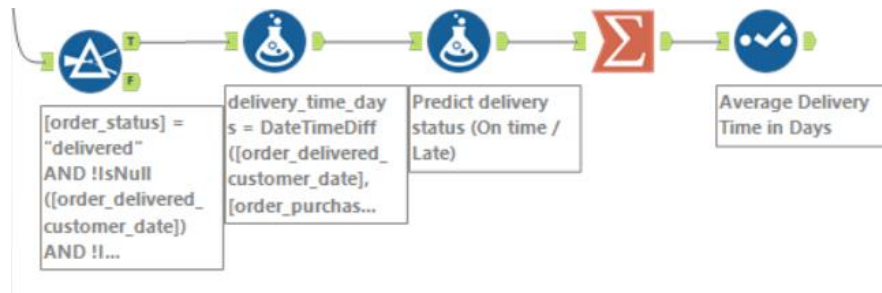


Figure 5.5.10 Delivery Performance Analysis Workflow

In order to get a better understanding of the logistics efficiency and customer satisfaction in the Olist market, customer orders are divided into two categories according to the estimated delivery time which are “on time” and “late”. As shown in Figure 5.5.11, 101475 orders were delivered within the expected time, with the average delivery time of 10.39 days, reflecting the overall stability and predictable logistics performance. In the other hand, 8714 orders were delayed and the average delivery time increased significantly to 30.89 days. This apparent gap indicates that when delays occur, they are usually driven by deeper supply chain disruptions within the regional delivery network rather than a slight scheduling deviation.

	delivery_status	Avg_delivery_time_days	Count
1	On time	10.386174	101475
2	Late	30.885931	8714

Figure 5.5.11 Distribution of Customer Orders by Delivery Status and Average Delivery Time

5.5.6 Customer Spending Segmentation

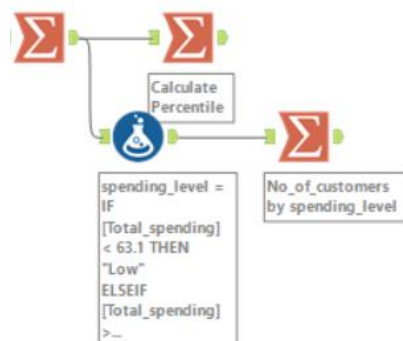


Figure 5.5.12 Customer Segmentation Workflow

To get a better understanding of customer purchasing behavior within the Olist marketplace, customers are divided into three spending levels which are low, medium and high based on the quartile analysis. As shown in figure 5.5.13, 23821 customers are classified as “Low” spenders, 47745 as “Medium” spenders and 23854 as “High” spenders. The distribution shows that most customers fall within the medium spending category while the remaining customers are evenly falling between the low and high spending category.

	spending_level	No_of_customers
1	Medium	47745
2	Low	23821
3	High	23854

Figure 5.5.13 Distribution of Customers by Spending Level

6.0 Challenges Faced

In the process of developing business intelligence solutions in Alteryx Designer, several technical challenges were encountered. One of the main challenges is data integration. Due to the lack of compatible keys in some tables, the originally selected dataset cannot be connected correctly. This led to incomplete connection and limits the effectiveness of the analysis process. In order to continue the project, our team migrated to the Olist e-commerce dataset, which provides a more structured relationship model for data integration.

Another challenge occurred in the data conversion stage when the output of the formula was accidentally assigned to the date field. This causes the original chronological information to be overwritten and affects the forecasting workflow. The problem was solved by storing the converted value in a separate field while retaining the original date column.

Our team also encountered difficulties in the prediction stage. Due to date anomalies and insufficient historical records, the initial prediction model produced unrealistic predictions. Therefore, the forecast does not accurately reflect the actual sales trend. In order to improve the performance of the model, the dataset is filtered to delete problematic records and keep a more consistent analysis time.

7.0 Dashboard Overview

This analysis presents two comprehensive Power BI dashboards with interactive visualizations and storyboards designed to provide actionable insights into sales performance, product category analysis, logistics efficiency and geographic distribution of operations.

7.1 Dashboard 1: Sales & Product Performance Dashboard

The main dashboard displays comprehensive KPI metrics and visualizations of sales trends, product performance, geographic distribution, and customer behavior.

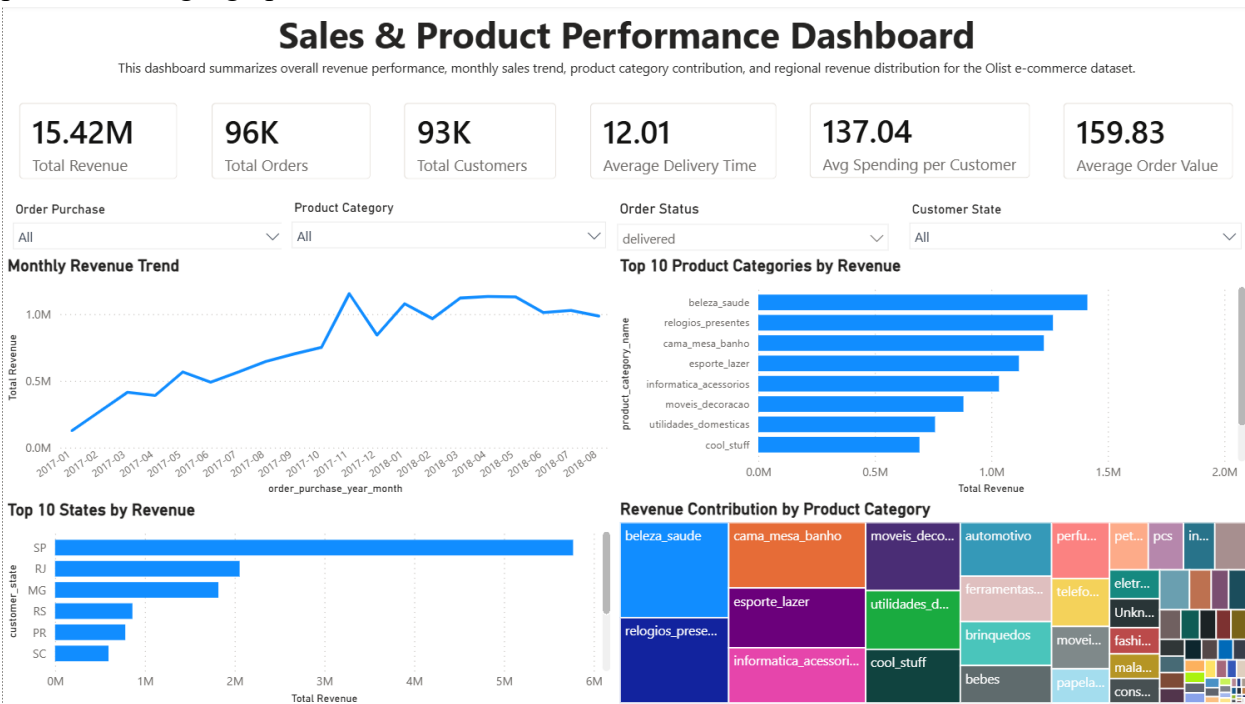


Figure 7.1: Sales & Product Performance Dashboard

Key Metrics Displayed:

- Total Revenue: \$15.42M
- Total Orders: 96K
- Total Customers: 93K
- Average Delivery Time: 12.01 days
- Avg Spending per Customer: \$137.04
- Average Order Value: \$159.83

7.1.1 Chart Choice Optimization & Visual Best Practices

1. KPI card (Total Revenue, Totals Orders, Total Customers, Average Delivery Time, Average Spending per Customer, Average Order Value)

- **Goal:** To establish a macro-level business baseline and present key performance indicators (KPI) immediately.
- **Chart selection & Justification:** Simple numeric cards are chosen to eliminate visual cutters and ensure that executive stakeholders can immediately grasp the core business health indicators without cognitive load.
- **Best Practice & Interactive:** Pair with large font with the concise category label below. When any slicer is cross filtered, these cards will dynamically change their values and act as a real-time reactive scoreboard.

2. Monthly Revenue Trend (Line Chart)

- **Goal:** To track the fluctuation of revenue over time and determine the historical growth pattern and seasonal demand peaks.
- **Chart selection & Justification:** Line chart is the standard best practice for continuous time series data. It effectively emphasizes the trajectory and rate of change of data, not just a single different point.
- **Best Practice & Interactive:** The chronological X-axis sorting can ensure the appropriate timeline flow. Clicking on the data point of a specific month will cross-filter the rest of dashboard 1 to show the top products and geographical revenue distribution of that specific month.

3. Top 10 Product Categories by Revenue (Horizontal Bar Chart)

- **Goal:** Rank and compare the total revenue generated by each product category to identify market leaders.
- **Chart selection & Justification:** Choose horizontal bar charts instead of vertical bar charts, because it provides clean and untruncated space for long product category names and improves readability.
- **Best Practice & Interactive:** Sort in descending order to enforce instant ranking understanding. Click the product bar to isolate and filter geographical and monthly trends to show the specific market performance of the project.

4. Top 10 States by Revenue (Horizontal Bar Chart)

- **Goal:** Expose the strength of the regional market and determine the dominant geographical area.
- **Chart selection & Justification:** Like product charts, the horizontal layout allows for a clear comparison of regional codes with minimal visual friction.
- **Best Practice & Interactive:** Apply consistent blue coding in major income indicators to maintain visual harmony. After clicking, it acts as an area filter for the entire dashboard.

5. Revenue Contribution by Product Category (Tree Map)

- **Goal:** Explain the relationship between the part and the whole and show the quantitative contribution of all product categories simultaneously.
- **Chart selection & Justification:** Tree charts handle a large number of categories better than pie charts. They use nested rectangles to display the proportions cleanly and do not overlap text.
- **Best Practice & Interactive:** The largest category is automatically positioned in the upper left corner for intuitive reading habits. It allows stakeholders to quickly see the diversification and concentration risks of the entire portfolio.

7.2 Dashboard 2: Logistics & Geography Dashboard

The logistics dashboard provides an overview of order fulfillment, delivery status, geographic customer distribution, and state-level performance metrics.

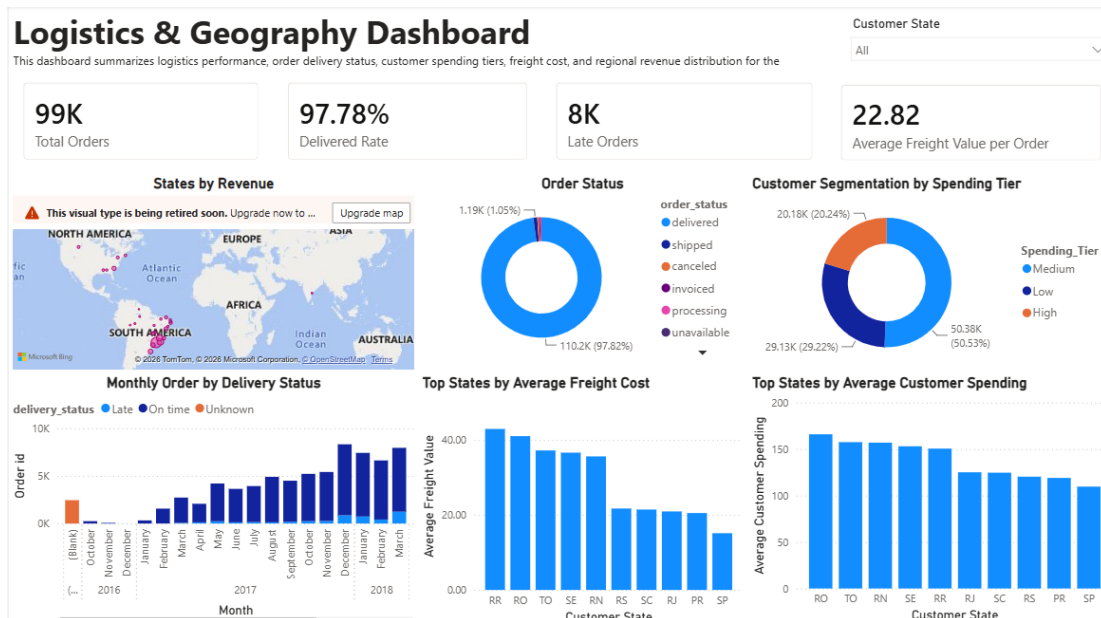


Figure 7.2: Logistics & Geography Dashboard

Key Metrics Displayed:

- Total Orders: 99K
- Delivered Rate: 97.78%
- Late Orders: 8K
- Average Freight Value per Order: \$22.82

7.2.1 Chart Choice Optimization & Visual Best Practices

1. KPI card (Total Orders, Delivery Rate, Late Orders, Average Freight Value per Order)

- **Goal:** To establish a macro-level business baseline and present key performance indicators (KPI) immediately.
- **Chart selection & Justification:** Simple numeric cards are chosen to eliminate visual cutters and ensure that executive stakeholders can immediately grasp the core business health indicators without cognitive load.
- **Best Practice & Interactive:** Pair with large font with the concise category label below. When any slicer is cross filtered, these cards will dynamically change their values and act as a real-time reactive scoreboard.

2. States by Revenue (Map Visualization)

- **Goal:** To project regional sales performance to the actual geographical boundaries to help logistics and regional expansion decision-making.
- **Chart selection & Justification:** Map vision uses spatial intuition to allow regional managers to immediately see the density and market penetration of Brazil's vast territory.
- **Best Practice & Interactive:** The size of the bubble directly corresponds to the amount of income, minimizing cognitive efforts. Users can zoom in/out and hover to view localization details and seamlessly cross-filter freight and expenditure indicators.

3. Order Status Breakdown (Donut Chart)

- **Goal:** To visualize the classification composition of all orders and verify the performance status ratio.
- **Chart selection & Justification:** The donut chart is chosen because the category is highly limited, making the proportion easy to distinguish, while keeping the center open and the layout clean.

- **Best Practice & Interactive:** High-contrast color selection separates the main states. Click on the status slice to isolate the delivery performance trend on the time chart.

4. Customer Segmentation by Spending Tier (Donut Chart)

- **Goal:** To analyze the percentage distribution of low, medium, and high consumption groups in active consumer groups.
- **Chart selection & Justification:** Like state subdivision, the donut chart neatly presents the partial-to-whole relationship of three different quartiles without confusing the canvas.
- **Best Practice & Interactive:** Data labels clearly show the exact number of employees and percentage contributions. Selecting a tier like "high-spending customers" will immediately highlight where the customers are geographically clustered and how much they contribute to each local state's average.

5. Monthly Order by Delivery Status (Stacked Column Chart)

- **Goal:** To review the stability of the supply chain over time by tracking the monthly "on time" and "late" deliveries.
- **Chart selection & Justification:** The stacked bar chart shows the absolute order volume of each month, as well as the internal proportional changes of delivery failures in the month.
- **Best Practice & Interactive:** Use clear color legend definitions to distinguish performance accuracy. If the order surge is related to the unacceptable surge of delayed delivery, it will immediately alert the management.

6. Top 10 States by Average Freight Cost (Clustered Column Chart)

- **Goal:** To conduct benchmark tests on the operating and transportation costs of different geographical states and isolate expensive logistics areas.
- **Chart selection & Justification:** Vertical bar charts are ideal for displaying discrete geographical categories to easily compare individual heights with overall baselines.
- **Best Practice & Interactive:** Sort in descending order and automatically push the most expensive average freight cost to the left so that it can be seen immediately.

7. Top 10 States by Average Customer Spending (Clustered Column Chart)

- **Goal:** To evaluate the business value and purchasing power of consumers in different geographical countries.
- **Chart selection & Justification:** Vertical bar charts provide a perfectly standardized baseline for comparing regional expenditure differences.
- **Best Practice & Interactive:** Data alignment follows the same visual hierarchy as freight charts, which can make multi-dimensional comparisons of regional profit margins.

8.0 Key Insight

8.1 Revenue Performance & Growth Trends

The organization has achieved a total revenue of \$15.42M across 96,000 orders from 93,000 customers. Highest revenue in November at approximately \$ 1.15M, driven by seasonal shopping demand. The lowest revenue was recorded in January 2017 at approximately \$127.4K, indicating a strong year-end purchasing effect. The average order value of \$159.83 demonstrates stable customer purchasing power and efficient order management.

8.2 Product Category Performance & Market Concentration

The Top 10 Product Categories analysis shows that 'beleza_saude' (Beauty & Health) which contribute about \$1.4M is the leading revenue generator, followed by 'religios_presentes' (Religious Gifts) with \$1.26M and 'cama_mesa_banho' (Bedding & Bath) with \$1.23M. This indicates significant market concentration in lifestyle and wellness categories. The product portfolio demonstrates diversification across multiple categories, reducing business risk while allowing for target marketing strategies.

8.3 Geographic Revenue Distribution & Regional Market Strength

The geographic analysis reveals that Sao Paulo (SP) is the dominant market, generating the highest revenue, followed by Rio de Janeiro (RJ) and Minas Gerais (MG). This concentration in major metropolitan areas suggests strong market penetration in densely populated regions. The presence of revenue across multiple states indicates geographic diversification, presenting opportunities for expansion into underperforming regions.

8.4 Delivery Performance & Logistics Efficiency

The logistics dashboard shows an impressive delivered rate of 97.78%, indicating excellent order fulfillment capability. However, 8,000 late orders represent a notable concern requiring attention. The

average delivery time of 12.01 days suggests a reasonable fulfillment window, though optimization opportunities exist. The average freight value per order of \$22.82 indicates moderate logistics costs, but regional variations suggest potential for cost optimization in higher-cost regions.

8.5 Customer Segmentation & Spending Behavior

The customer segmentation analysis reveals three distinct spending tiers: High-value customers (56,444 customers, 50.1%), Medium-tier customers (34,644 customers, 30.75%), and Low-value customers (21,576 customers, 19.15%). This distribution indicates healthy customer base composition with significantly high-value customer concentration. The average customer spending of \$137.04 reflects solid customer lifetime value.

9.0 Business Recommendations

9.1 Optimize Delivery Performance & Reduce Late Orders

While the 97.78% delivery rate is commendable, the 8,000 late orders impact customer satisfaction and operational efficiency. Recommendations:

- Conduct root cause analysis on late orders to identify bottlenecks
- Implement real-time tracking and proactive customer notification systems
- Negotiate with logistics partners to reduce delivery times to under 10 days
- Target: Reduce late orders to below 5% within 6 months

9.2 Expand High-Performing Product Categories

Leverage the strong performance of beauty & health and religious gifts categories:

- Increase inventory in top 5 product categories contributing 60% of revenue
- Launch seasonal campaigns targeting peak demand periods
- Develop cross-selling strategies between complementary categories
- Target: Increase average order value by 15% annually

9.3 Geographic Market Expansion & Regional Development

Develop strategies to penetrate underperforming regions:

- Analyze regional demand patterns and tailor product offerings
- Establish regional distribution centers in high-demand states
- Implement targeted digital marketing campaigns for emerging markets

- Target: Achieve 25% revenue growth from secondary markets within 12 months

9.4 Optimize Logistics Costs & Regional Performance

Address regional freight cost variations:

- Analyze high-freight-cost states and negotiate better rates
- Implement regional logistics hubs to optimize distribution networks
- Target: Reduce average freight cost by 10-15% through optimization

9.5 Customer Segmentation Strategy & Retention Focus

Develop tailored strategies for each customer segment:

- Implement loyalty programs targeting high-value customers
- Launch upselling campaigns to move medium-tier customers to high-value status
- Design re-engagement campaigns for low-value customers
- Target: Increase average customer spending from \$137.04 by 20%

10.0 Discussions

10.1 Overall Business Performance

The dashboard results show that the Olist e-commerce dataset generated a total revenue of \$15.42 million from approximately 96,000 orders and 93,000 customers. The average order value was \$159.83, while the average spending per customer was \$137.04. These values indicate that the platform has a strong transaction volume and stable customer purchasing power. However, since the average spending per customer is lower than the average order value, it suggests that many customers may only purchase once or make limited repeat purchases. Therefore, the business should focus not only on acquiring new customers, but also on increasing customer retention and repeat purchase behavior.

10.2 Product and Regional Performance

The product category analysis shows that beauty and health contributed the highest revenue at approximately \$1.4 million, followed by religious gifts with \$1.26 million and bedding and bath with \$1.23 million. This shows that revenue is strongly concentrated in lifestyle, personal care, and household-related product categories. From a business perspective, these categories should receive higher priority in

inventory planning, seasonal promotion, and seller partnership strategies. At the same time, the geographic analysis shows that São Paulo, Rio de Janeiro, and Minas Gerais are the strongest revenue-contributing states. This indicates that the business performs better in major metropolitan regions, while weaker states may still have potential for future market expansion.

10.3 Logistics and Customer Behavior

The logistics show that the delivery rate reached 97.78%, which indicates that most orders were successfully fulfilled. However, there were still around 8,000 late orders, and the average delivery time was 12.01 days. In the detailed Alteryx output, on-time orders had an average delivery time of 10.39 days, while late orders increased significantly to 30.89 days. This large difference shows that late delivery is not a small delay issue, but a serious logistics performance gap. Therefore, delivery performance should be monitored using late order count, average delivery days, and freight value instead of only looking at delivered rate.

10.4 Limitations and Improvement

Although the BI solution provides useful business insights, there are several limitations. First, the forecasting analysis was based on a limited historical monthly data range, so the forecast result should be treated as a supporting insight rather than a fully conclusive prediction. Second, the dataset mainly contains transaction, product, customer, and delivery information, but it does not include customer reviews, seller ratings, marketing campaigns, competitor pricing, or economic factors. These missing external factors may influence customer purchasing behavior and delivery satisfaction. Future improvements can include more data sources and more advanced dashboard features such as drill-through pages, dynamic customer segmentation, and predictive monitoring for late delivery risk.

11.0 Conclusions

In conclusion, this project developed an end-to-end business intelligence solution for analyzing Brazilian e-commerce customer purchasing behavior using the Olist datasets. Alteryx Designer was used to perform data input, cleaning, transformation, integration, and analytical output generation, while Microsoft Power BI was used to create interactive dashboards and visual storytelling. The final dashboards successfully presented key business areas such as total revenue, order performance, product category contribution, regional revenue distribution, delivery efficiency, and customer spending segmentation.

The analysis showed that the business achieved strong revenue performance, with major contributions from selected product categories and high-performing states. At the same time, the logistics dashboard

highlighted the importance of reducing late orders and optimizing freight cost across regions. Overall, the project proved that a structured BI workflow could support better decision-making by converting raw operational data into clear, interactive, and actionable business insights.

12.0 References

1. Olist. (n.d.). *Brazilian e-commerce public dataset by Olist*. Kaggle. Retrieved May 11, 2026, from <https://www.kaggle.com/datasets/olistbr/brazilian-e-commerce>

13.0 Appendix

13.1 Alteryx Workflow

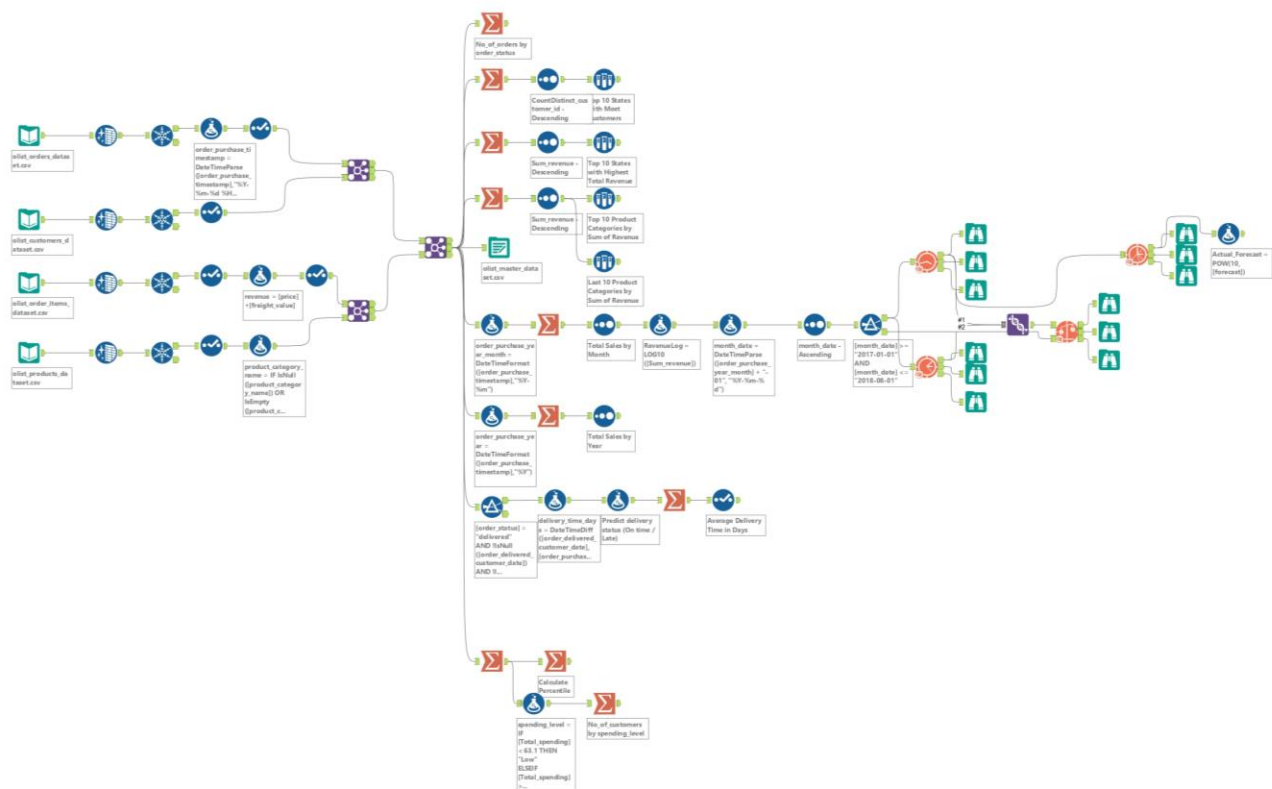


Figure 13.1.1: Alteryx Workflow Overview